

Experiment report

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Course: Introduction to Big Data Technology

Experiment Topic: Web Crawler

Task Description

Chosen website: <https://store.steampowered.com/>

How much data I want to download: five pages

Experiment Code

1. Import libraries

```
import urllib.request
from bs4 import BeautifulSoup
import requests
```

2. The User-Agent header is specified to prevent the server from blocking the request, as some websites restrict access based on the client type. The new target URL is the Steam Store homepage. The response from the server is stored in response, and its content (HTML) is accessed via response.text. Then the HTML content of the Steam Store page is printed to the console.

```
3 headers = {'User-Agent': 'Mozilla/5.0 (Windows NT 10.0; WOW64) AppleWebKit/537.36 (KHTML, like Gecko) '
4           'Chrome/51.0.2704.63 Safari/537.36'}
5 url="https://store.steampowered.com/"
6 response=requests.get(url,headers=headers)
7 data=response.text
8 print(data)
```

3. Then I used the BeautifulSoup library to parse the selected url and then I'm finding the headers which contain links. Then checking the condition (I need only 5 links) and add them to the list.

```
soup = BeautifulSoup(data, features='html.parser')
header_links = []
for h2 in soup.find_all('h2'):
    a_tag = h2.find('a')
    if a_tag and len(header_links) < 5:
        header_links.append((h2, a_tag['href']))
```

4. Then I'm creating a `html_content` element to save the `header_links` that I fetch using web crawler.

```
20 html_content = "<html><head><title>Crawler</title>5 links:</head><body>"
21 html_content += "<h1></h1><ul>"
22
23 for h2, link in header_links:
24     html_content += f"<li><a href='{link}'>{h2.text}</a></li>"
25
26 html_content += "</ul></body></html>"
```

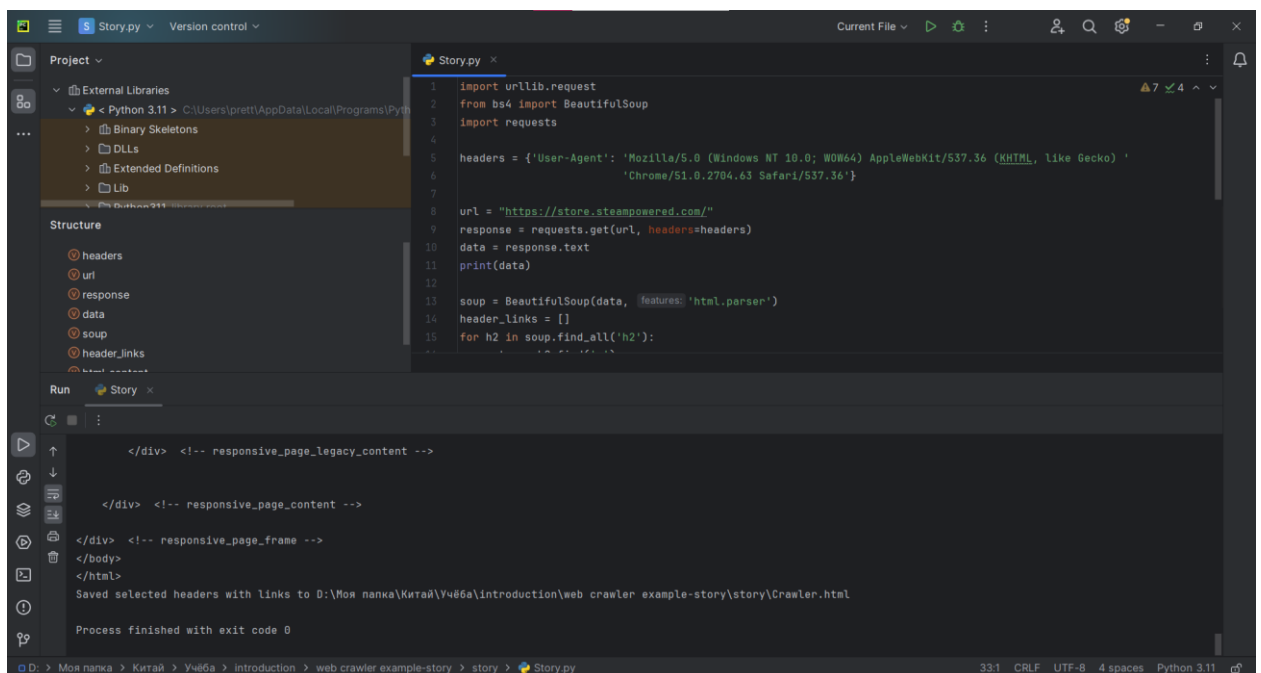
5. I changed the file path to "D:\Моя папка\Китай\Учёба\introduction\web crawler example-story\story\Crawler.html".

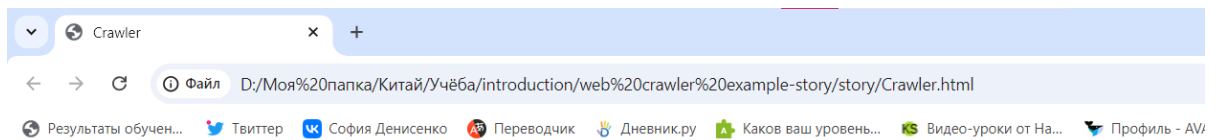
The `open()` function is used to open the file for writing. And it writes data to the file.

And printing the success message to the console.

```
28 file_path = "D:\Моя папка\Китай\Учёба\introduction\web crawler example-story\story\Crawler.html"
29 with open(file_path, "w", encoding="utf-8") as f:
30     f.write(html_content)
31
32 print(f"Saved selected headers with links to {file_path}")
```

Result





5 links:

- [Special Offers Browse More More](#)
- [Recommended based on the games you play More](#)
- [Your Discovery Queue Learn More](#)
- [Customize, Explore by Tag, & More More These games today](#)
- [The Community Recommends More](#)

